Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda (remote, UTC+3)

PROFESSIONAL SUMMARY

Software engineer with 6+ years of fully remote work on **real-time, asynchronous and stateful systems** in Node.js, with Python and TypeScript in production. Designed a live two-way sync between MongoDB and Firestore on Google Pub/Sub and used it to migrate 20+ production apps with **zero downtime**. Built a USSD channel over gRPC for users on unreliable networks. Builds **LLM-based applications** outside work: an OpenAI + Vision assistant over WhatsApp with conversation state, and an **agentic pipeline** on Claude Code that checks every source before a model sees it. Wrote SDK documentation for other engineers. Now applying the same state and concurrency work to **agent frameworks and developer APIs**. Code: github.com/zac-09.

CORE COMPETENCIES

State, Concurrency & Async Workflows | Real-Time Sync (Pub/Sub, Kafka, NATS) | LLM Applications & Agentic Workflows | Conversational AI (Messaging Channels) | Node.js, Python & TypeScript | Distributed Systems & gRPC | Developer Docs & Enablement | Remote Collaboration

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Built a real-time **two-way sync** between MongoDB and Firestore with Node.js and **Google Pub/Sub** message processing. Both stores stayed consistent under live production traffic during the cutover.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built a customer-facing **3D visualisation feature** using video overlay and the **ffmpeg** library, increasing buyer conversion.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a **Python** script. Saved **$5,000/month** by moving all services off AWS.
- Shipped Express Shipping, generating an additional **$40 revenue per order**. Migrated a Webflow website from a Parse backend to Firebase.

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Built the migration script that read MongoDB data, ran complex processing logic and wrote to Firestore.
- Wrote **developer documentation** for fellow engineers on the new Firebase SDKs (Node.js and Python), and trained the Ops team on the new dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech PWA

- Extended backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js.
- Built ad-hoc jobs over **2M+ Firestore records** to keep real-time stock prices current from Morningstar APIs. Added responsive React and Tailwind pages to the PWA.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Built Node.js backends for DeFi applications using Web3, and delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech

- Architected an **event-driven microservices** backend for FIDA Uganda's case management app on **Apache Kafka**, Docker and Kubernetes.
- Built a **USSD service** that let legal aid providers reach a Node.js backend over **gRPC**.
- Led a team of 4 developers. Built a paralegal database for LASPNET Uganda and a React Native + Expo app (iOS & Android) with Firebase push notifications.

PROJECTS

Healthcare WhatsApp Assistant LLM · text channel

A conversational assistant inside WhatsApp that keeps conversation state, accepts prescription photos and replies in the chat. Uses OpenAI for dialogue and Google Vision for OCR, behind the WhatsApp Business API.

OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

Agentic Evaluation Pipeline agents

Extended an open-source agentic pipeline on Claude Code. It scans job portals, checks each source live in a headless browser before any model scores it, deduplicates inputs and drafts tailored documents. A human approves every submission. 200+ evaluations run so far.

Claude Code · Playwright · Node.js · YAML

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, Python, TypeScript, HTML, CSS, Go (intermediate), Rust (intermediate)

AI: OpenAI API, Google Cloud Vision, Claude Code agentic workflows

Async & Messaging: Google Pub/Sub, Apache Kafka, NATS Streaming, gRPC, USSD

Backend & Data: Node.js, Express.js, Firebase Cloud Functions, MongoDB, Firestore

Cloud / Infra: GCP, Firebase, AWS (intermediate), Docker, Kubernetes, Linux, Git

Media / Frontend: ffmpeg (video overlay), React, React Native (intermediate), Tailwind **Spoken:** English, Kiswahili